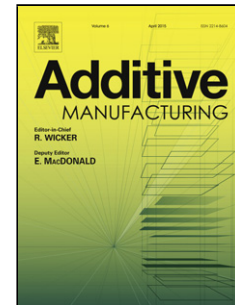


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# Parametric Studies on Bending Stiffness and Damping Ratio of Sandwich Structures

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## Abstract

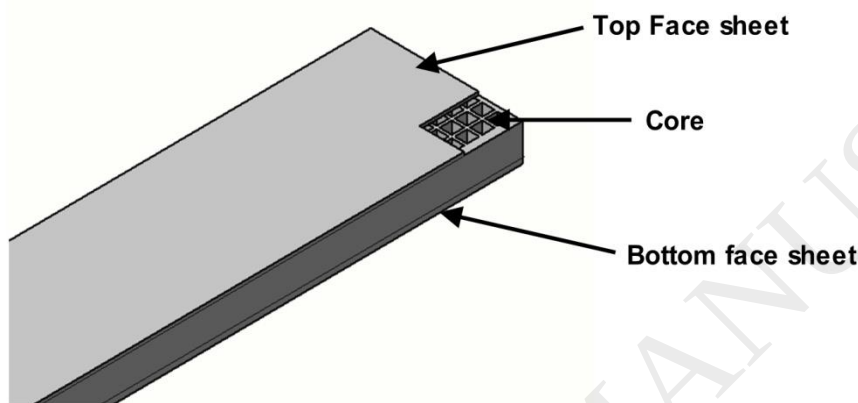
Sandwich structures are extensively used in aviation industries to reduce the overall weight of the system. Although the mechanical behavior of these structures has been widely studied, the performance of core shape in vibration response has been minimally explored. This study focuses on understanding the various influences of sandwich structures considering the following parameters: (i) nature of core shape, (ii) number of infill shapes, and (iii) orientation of cores, which affect the dynamic behavior namely of sandwich structures. Nine sandwich structures comprising three different core shapes, hexagon, triangle, and square shapes, in three different orientations, namely  $0^\circ$ ,  $45^\circ$ , and  $90^\circ$ , were considered for the present study. These structures in the beginning were put by modal analysis using finite element method (FEM). All the nine structures were printed using the fused deposition method to validate the FEM findings, while the DEWE soft data acquisition system was used to estimate the modal parameters (i) natural frequency and (ii) damping ratio. Natural frequency and damping ratio were estimated using FRF and Nyquist circle plot, respectively. This study demonstrates that although the square core orientated at  $0^\circ$  exhibited superior stiffness in bending loads, the hexagonal core orientated at  $0^\circ$  displayed an admirable combination of both stiffness and damping properties.

**Keywords:** Sandwich beam, additive manufacturing, sandwich cores, vibration, modal analysis

## 1. Introduction

The demand for new materials and structures results from the vigorous efforts made by the automobile and aviation industries for saving the natural resources (fuel) and

concurrently reducing the emission of greenhouse gases. Hence, conventional solid structures will make way for lightweight ones without compromising on their mechanical/structural properties in the desired direction. A typical sandwich structure containing a top and bottom face sheets separated by a relatively lightweight core, as shown in Fig. 1, is considered to be one of the most desirable solutions[1]. Sandwich structures provide a high stiffness to weight ratio, energy absorption capacity in the desired direction[2], and efficient transmission and absorption of acoustic noise[3]. Nevertheless, the static and dynamic mechanical behavior of a sandwich structure depends on the material, geometry of the face sheets and cores[4].



**Fig. 1** Schematic representation of a sandwich structure

In sandwich structures, the face sheets are generally thin, stiff, and strong compared to the thick and soft-core structure. The core structure is made of honeycomb, triangular truss members etc. and its desired functions are as follows: (i) increase bending and buckling resistance [5], (ii) increase shear stiffness and energy absorption ability [6,7], and (iii) provide enhanced static and dynamic properties. Designating a core having superior dynamic properties may enhance the performance of applications such as cabins of washing machines, automobile, and aircraft [8–10]. The selection of a proper infill core for an application can reduce the occurrence and impact during resonance in the system by making the system stiffer. Furthermore, the effectiveness of the sandwich structure with superlative cores can be enhanced by filling the smart materials into a specific area to obtain desired dynamic properties using an external stimulus such as magnetic field [11–16].

Furthermore, the core's infill can be used in the 3D printer to substitute the solid structure, which employs minimal material, preserves strength, and saves manufacturing time. Nevertheless, it is essential to establish and understand the mechanical/structural dynamic behavior of sandwich structures before implementing them in their intended

applications. In this regard, various researchers have focused their studies on understanding the properties of sandwich structures with different core shapes and sizes. Table 1 consolidates the findings by various researchers. In this regard, Wang and Quek [17] developed a mechanics model for the vibration analysis of a sandwich beam with a piezoelectric layer, and they envisaged that the dynamic characteristic of the entire structure was correlated to the position of the piezoelectric layer. Gordaninejad and Bert [18] developed a model to investigate the bending of sandwich beams with relatively thick facings and moderately stiff cores. In their modeling, they included transverse shear deformation in the face sheets along with stretching and bending deformation in the core. Hyer *et al.*[19] have investigated the nonlinear vibration of a sandwich beam with a viscoelastic core. Yurddaskal *et al.* [20] performed numerical and experimental analysis to determine the effects of curvature and foam properties on the natural frequencies of the sandwich panels. They concluded that the natural frequencies increased with the increase in curvature and foam density; however, the increase due to an increase in the magnitude of curvature decreases with the increase in foam density. Frostig [21] investigated the behavior of mid-plane asymmetric sandwich beams with a flexible core. In the aforementioned studies, researchers have focused their studies on studying the dynamic properties of the sandwich beam as a whole. Further, researchers[1,4,8,12,22–30] have focused their studies on understanding the dynamic properties of sandwich structures with respect to core shapes, size, and their orientation, and their findings are consolidated in Table 1.

**Table 1** Existing literature findings to understand the nature of core shapes in sandwich structures.

| Author/Ref. No.               | Shape or orientation of Core/fiber         | Evaluation Properties                               | Findings                                                                                                                                                          |
|-------------------------------|--------------------------------------------|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Zaid <i>et al.</i> [1]        | Trapezoidal                                | Compression loading                                 | Compression loading is sensitive to the number of unit cells and the core thickness.                                                                              |
| Li <i>et al.</i> [4]          | Triangular, hexagonal, and star-cell shape | Young's modulus, shear modulus, and Poisson's ratio | All three cores had the same stiffness, and the triangular core had lower compressive and shear buckling strength. The star-cell core exhibited high flexibility. |
| Arunkumar <i>et al.</i> [12]  | Honeycomb, Triangular                      | Sound transmission loss                             | Core face sheet thickness significantly affects the sound transmission loss.                                                                                      |
| Jeom <i>et al.</i> [31]       | Honeycomb                                  | Buckling, collapse, and crushing behavior           | Overall thickness, height, and wall thickness of the core affect the mechanical properties of the honeycomb structure.                                            |
| Dharmasena <i>et al.</i> [22] | Square                                     | Mechanical response during explosive tests          | Square core structure suffered less back face deflections compared to solid plates of identical mass.                                                             |
| Zaldivar <i>et al.</i>        | Laminated core                             | Tensile strength, failure                           | All properties varied significantly with                                                                                                                          |

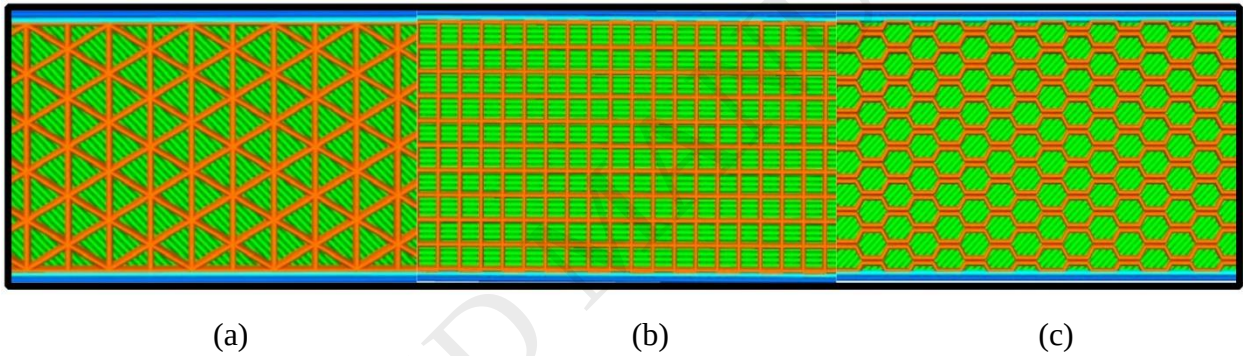
|                                       |                                                   |                                                                                |                                                                                                                                                                                                                                                       |
|---------------------------------------|---------------------------------------------------|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>al. [23]</i>                       | using 3D Printing                                 | strain, Poisson's ratio, coefficient of thermal expansion, and elastic modulus | the build orientation, and hence, the build orientation decides the strength required in a particular direction.                                                                                                                                      |
| Melenka <i>et al. [24]</i>            | Continuous fiber-reinforced 3D printed structures | Tensile properties                                                             | Increase in the volume of fiber reinforcement resulted in the increase in the stiffness and ultimate strength of the test samples.                                                                                                                    |
| Fleck <i>et al. [25]</i>              | Pyramid, diamond, corrugated, hexagon, square     | Blast resistance                                                               | Diamond-celled core provides the best blast performance.                                                                                                                                                                                              |
| Fernandez-Vicente <i>et al. [26]</i>  | Laminates printing orientation 0°, 45°, and 90°   | Stiffness                                                                      | Density, orientation, and their combination affect the stiffness of the manufactured parts.                                                                                                                                                           |
| Bak <i>et al. [27]</i>                | Honeycomb (20%), rectilinear (20, 50, and 100%)   | Tensile strength and stiffness                                                 | Rectilinear pattern in a 100% infill showed the highest tensile strength. For the same density, the honeycomb pattern showed a better tensile strength in less than 5% infill.                                                                        |
| Shohei Higashiyama <i>et al. [28]</i> | Honeycomb, square, and triangular                 | Heat dissipation and structural response                                       | Designed a compact and lightweight heat exchanger by a proper selection of core shape to have a better structural rigidity.                                                                                                                           |
| Hozhabr Mozafaria <i>et al. [29]</i>  | Honeycomb                                         | First natural frequency and mode shape                                         | Increase in the core thickness resulted in increase in the bending and torsional natural frequency; however, it decreases with the increase in the cell size of the core.                                                                             |
| A.A. Khdeir Professor [30]            | Softcore sandwich beam                            | Natural frequency and mode shapes                                              | Formulated a zigzag theory to conduct free vibration test by increasing the aspect ratio, i.e., the length to thickness ratio. Resonant frequency was found to be enhanced, while it decreased with the increase in the core to face thickness ratio. |
| Umberto Berardi [8]                   | Honeycomb                                         | Tri-axial vibration amplitude at the resonant frequency                        | Focus was on the computation of resonant frequency and the corresponding amplitudes of the water-stop tray of the washing machine. The central part of the tray was made stiffer by using honeycomb ribs.                                             |

From Table 1, it can be concluded that the mechanical/structural properties of the core depend on the following parameters: (i) shape of the structures such as honeycomb, square, and triangular, (ii) their orientation, (iii) density of infill, and (iv) dimensions of the core: thickness, height, wall thickness, etc. Hitherto, analytical and numerical analysis had been the focus of previous researchers leaving out the experimental analysis with the numerical investigation to determine the dynamics of sandwich structure in terms of the variation in core shapes, orientations, and their numbers. Moreover, the effect of core shapes on dynamic properties is yet to be studied and established, although there is understanding on the static mechanical properties of different core shapes. Therefore, the objective of the

present work is to understand and compare the dynamic properties of the sandwich structure with various infill core shapes so that a specific core shape can be effectively utilized for the required loading condition. In this regard, a beam consisting of (i) different core shapes (a) triangle, (b) square, and (c) hexagon/honeycomb is shown in Fig. 2(a, b, c), respectively, and (ii) different orientations of shapes (a)  $0^\circ$ , (b)  $45^\circ$ , and (c)  $90^\circ$  are studied by maintaining the same mass, thickness of face sheet, material, and outer dimensions.

A sandwich beam encompassed with the same outer dimensions can be characterized by either considering the same mass or the same volume. This fact can be considered in the present study, as the change in mass will influence the natural frequency of the system, and it will be difficult to interpret the results. In Eq. (1),  $\omega$  is the natural frequency,  $K$  is the stiffness of the structure, and  $m$  is the mass of the system.

$$\omega = \sqrt{\frac{k}{m}} \quad (1)$$

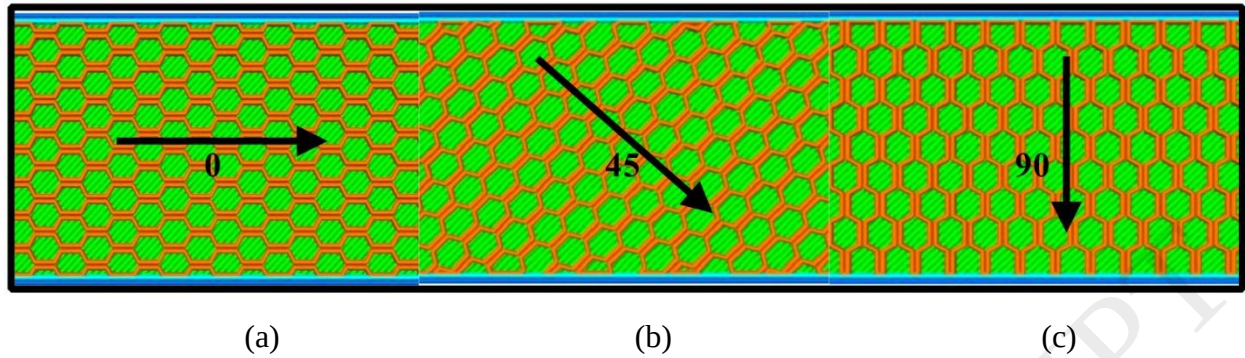


**Fig. 2** Shapes of core considered in the present work: (a) triangle, (b) square, and (c) hexagon

Given that the volume occupied by each core shape is different, it is expected that for given outer dimensions of a sandwich beam, the number of infills has to be different for different shapes and orientations to achieve the same mass. For illustration, different orientations (a)  $0^\circ$ , (b)  $45^\circ$ , and (c)  $90^\circ$  of honeycomb structures are shown in Figs. 3(a, b, c), respectively, and from these figures, it can be envisaged that the number of infills is different for different shapes.

In the present work, modal analysis [32] is performed initially for the different configurations of the sandwich structure using FEM by assuming fixed-free boundary condition. Structures with different core shapes and orientations are modelled in Autodesk Inventor and imported to ANSYS Workbench to perform the modal analysis. From the analysis, natural frequency and damping for different sandwich structures are determined and compared with experimental values. To validate the FEM results, all the nine sandwich

structures that have different core shapes and orientations are manufactured by the fused deposition method (FDM) with acrylonitrile butadiene styrene polymer (ABS) as a material.



**Fig. 3** Different orientations of the honeycomb structure used in this study: (a)  $0^\circ$ , (b)  $45^\circ$ , and (c)  $90^\circ$

FDM is a 3D printing technique, which is capable of manufacturing composites of any complex shape that may be difficult to manufacture by conventional manufacturing methods [27]. In FDM, the heated nozzle is translated into 2D space to produce one layer of the actual shape of the component. Layers are deposited over the previously developed layers by diffusion to build the full model. Applications of additive manufacturing include defense, medical equipment, aircraft, spacecraft, heat exchangers, and musical instruments [9,10,33–35].

Therefore, the present study attempts to affirm the core having superior dynamic properties developed by FDM. A theoretical investigation by FEM and experimental analysis on a developed setup is performed to find the effect of core shapes and orientations on the dynamic properties of developed sandwich structures. Finally, the sandwich structure that possesses superior stiffness and damping is proposed.

## 2. Experimental Setup

This section details the methodology followed for performing modal analysis by FEM and how various sandwich structures are developed using the 3D printer. Furthermore, the development of experimental setup and details of instruments used for evaluating the dynamic properties of structures are presented.

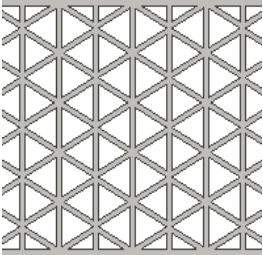
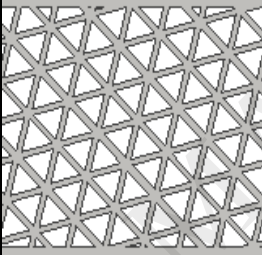
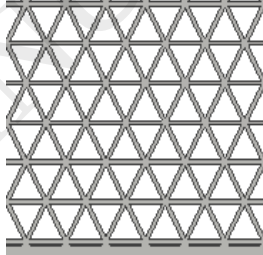
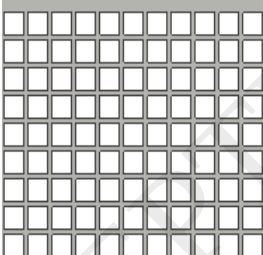
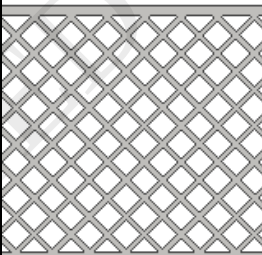
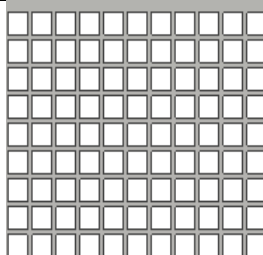
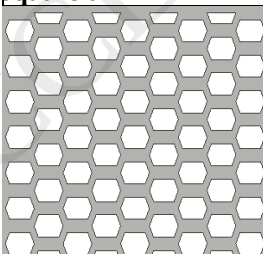
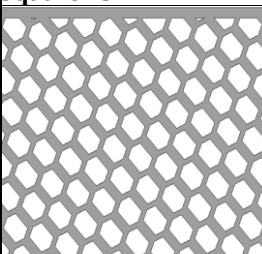
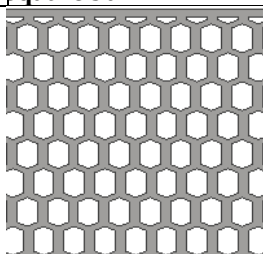
### 2.1. Modal analysis using finite element method

For performing modal analysis, sandwich beams containing different core shapes (a) triangle, (b) square, and (c) hexagon and different orientations (a)  $0^\circ$ , (b)  $45^\circ$ , and (c)  $90^\circ$ , of outer dimensions of length 160 mm, breadth 25 mm, and thickness 6 mm, are modelled in Autodesk Inventor. Because the natural frequency of the system depends on the mass and the stiffness of the sandwich structures, in the present work, the mass of the specimen is



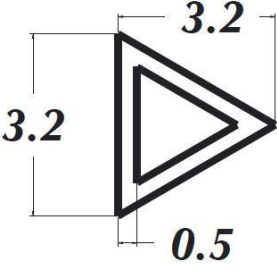
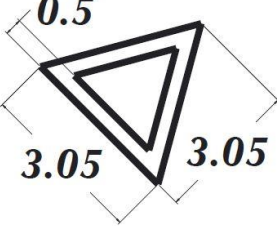
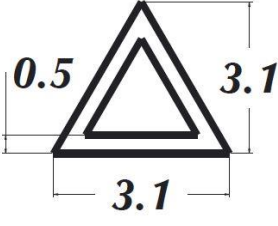
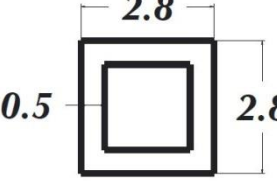
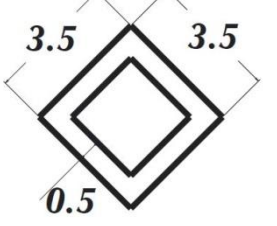
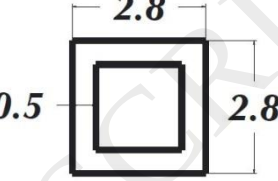
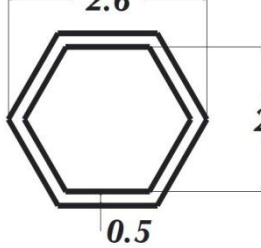
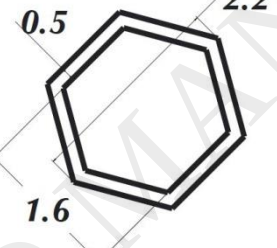
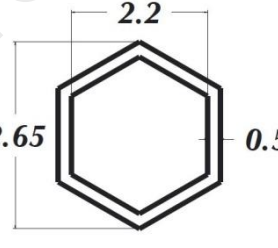
maintained the same by varying the infill percentage of cores. As a result, the variation in natural frequency can be studied with variation in the nature of core and the orientation of fiber. This method allows comparison of the number of core cells for providing better dynamic properties. The detailed sandwich structures are provided in Table 1. The modelled nine sandwich structures are shown in Table 2, and the dimensions of each core shape in millimeters are shown in Table 3. Designed model of these structures is imported in ANSYS Workbench 15, and modal analysis is performed by considering the structure as a cantilever beam.

**Table 2** Triangle, square, and hexagon-shaped specimens considered in the present work

|                                                                                     |                                                                                     |                                                                                      |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|   |                                                                                     |                                                                                      |
| <b>Sandwich structure</b>                                                           |                                                                                     |                                                                                      |
|   |   |   |
| <b>Triangular 0°</b>                                                                | <b>Triangular 45°</b>                                                               | <b>Triangular 90°</b>                                                                |
|  |  |  |
| <b>Square 0°</b>                                                                    | <b>Square 45°</b>                                                                   | <b>Square 90°</b>                                                                    |
|  |  |  |
| <b>Hexagon 0°</b>                                                                   | <b>Hexagon 45°</b>                                                                  | <b>Hexagon 90°</b>                                                                   |



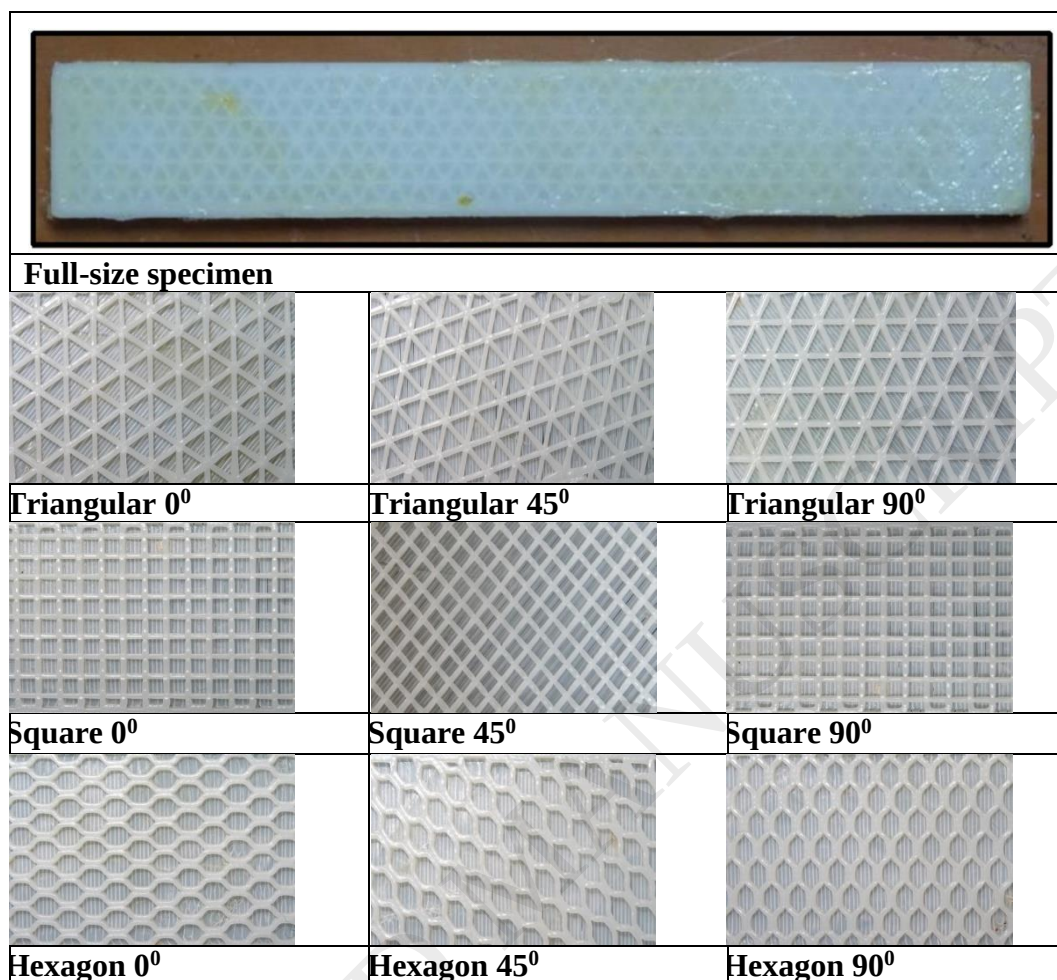
**Table 3** Dimensions of each core shape used in this study.

| Sandwich core shapes                                                               |                                                                                    |                                                                                      |
|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|   |   |   |
| Triangular 0°                                                                      | Triangular 45°                                                                     | Triangular 90°                                                                       |
|   |   |   |
| Square 0°                                                                          | Square 45°                                                                         | Square 90°                                                                           |
|  |  |  |
| Hexagon 0°                                                                         | Hexagon 45°                                                                        | Hexagon 90°                                                                          |

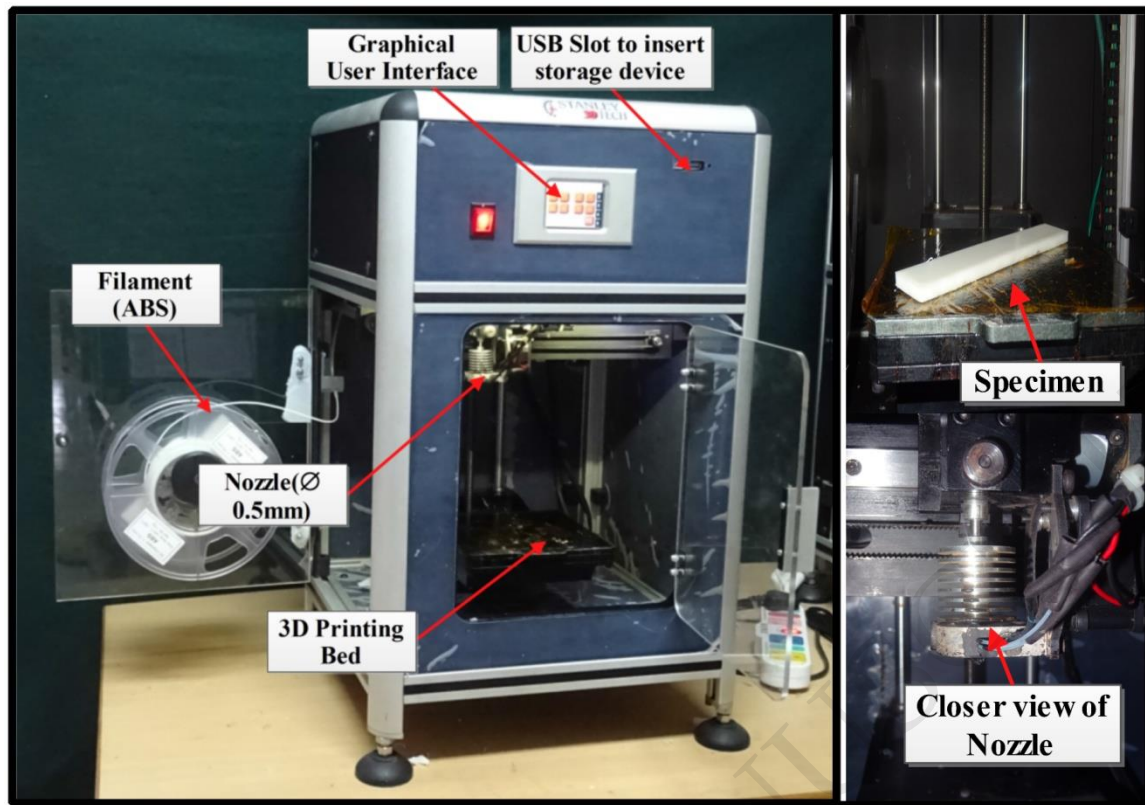
## 2.2. Development of specimens by FDM

Sandwich structures containing different core shapes and orientations are developed using Stanley Rheo 400 3D printer, as shown in Fig.4. 3D printers can be used to develop sandwich structures in two different directions: (i) vertical and (ii) horizontal. Components printed in vertical direction would require more number of layers to be diffused to achieve the final structure leading to increased printing time and reduced strength. Hence, horizontal position is chosen to print every specimen in this present research. Fig. 4 depicts the FDM-based printer with zoomed images of nozzle and printing bed.

**Table 4** shows triangle, square, and hexagon core shaped specimens studied in the present work



Specimens were modelled in Autodesk inventor modelling software and were fed to simplify 3D, the software that generates G codes to build the specimens. Acrylonitrile butadiene styrene (ABS) is chosen as the printing material. A raw filament of 1.75-mm diameter was fed to the nozzle that was heated to its melting temperature of 260°C, and to avoid warping in the printed component, the bed was heated to 100°C by using an induction heater. Specimens developed using the 3D printer is shown in Table 4.

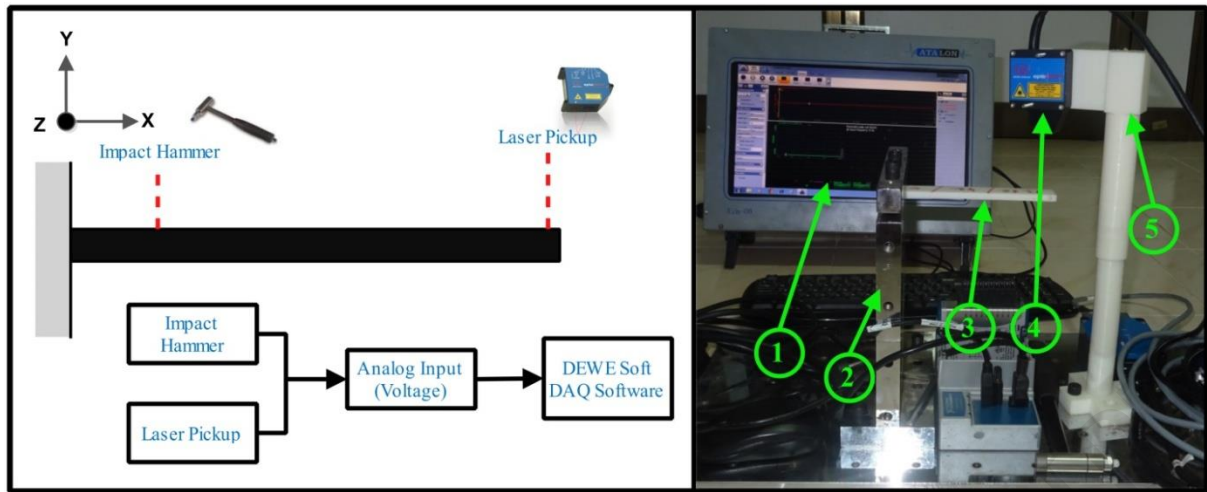


**Fig. 4** FDM-based 3D printing machine used in the study

### **2.3. Experimentation on the developed specimen**

In the present work, the experiments are performed by considering the beam as a cantilever one, i.e., fixing one end of the beam to a rigid fixture and allowing another end to vibrate freely.

To determine the frequency response function (FRF) of the specimen, the impact hammer test is performed using the DEWE soft data acquisition system as shown in Fig 5. X, Y, and Z axes shown in Fig. 5 are taken as a reference to represent the vibrations occurring in transverse, longitudinal, and bending directions, respectively. To avoid the additional mass effect on specimens, the response of the beam is measured using a noncontact-type laser displacement sensor. The laser displacement sensor is attached to a vertical stand, which is further connected to a rigid base. Laser pickup is powered by a 24-Volt DC power supply from a step-down voltage device.



**Fig. 5** Schematic representation and experimental setup with the hardware used in the study. (1) DEWE Soft Software, (2) Fixture, (3) Sandwich beam, (4) Laser pickup, (5) Laser pickup fixture.

The output voltage obtained from the laser displacement sensor is acquired using an analog input data acquisition (DAQ) unit [36]. FRF, phase diagram, and damping ratio of the beams were obtained using the DEWE soft data acquisition unit. Fig. 6 shows a typical output received from the DEWE system by conducting modal testing at five different nodes. The detailed explanation for estimating the resonant frequency and damping ratio by the DEWE Soft system is as follows:

The governing differential equation of a continuous system excited by a step input  $F(t)$  is written as follows:

$$M\ddot{x} + C\dot{x} + Kx = F(t) \quad (2)$$

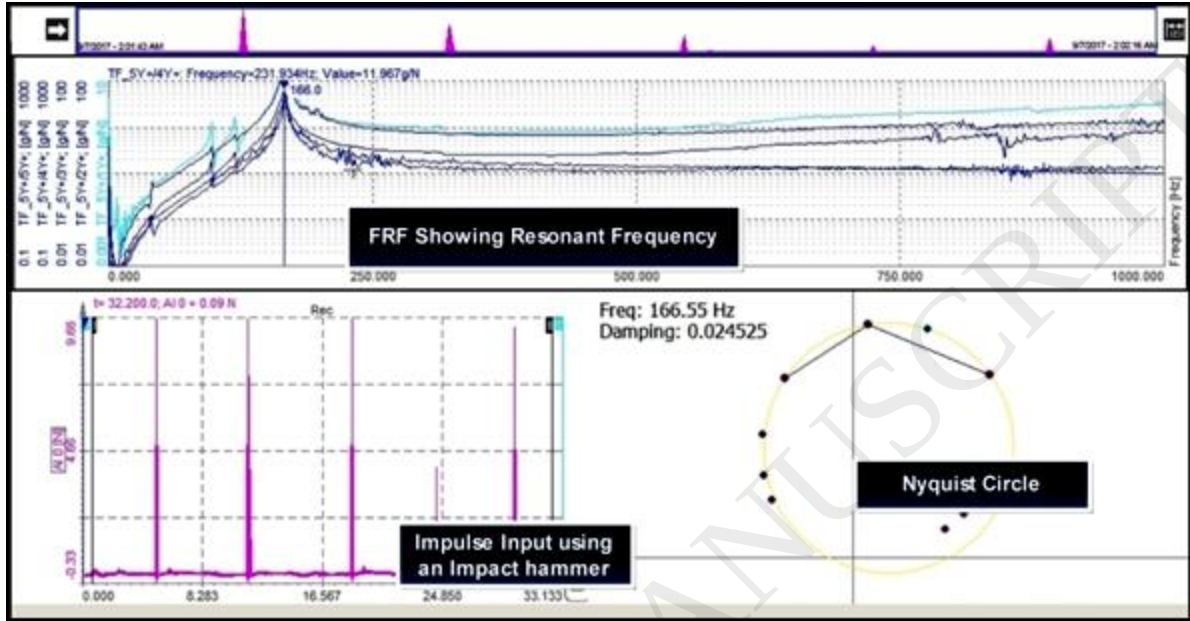
where  $M$ ,  $C$ , and  $K$  are the modal mass, damping, and stiffness of the continuous system, respectively. The receptance of the system is computed as follows [15, 32]:

$$H(j\omega) = \frac{X(j\omega)}{F(j\omega)} = \frac{1}{-\omega^2[M] + j\omega[C] + [K]} \quad (3)$$

where  $F(j\omega)$  represents the impulse input force imparted to the system using an impact hammer,  $X(j\omega)$  represents the response of the system that was measured by a noncontact-type sensor laser pickup and  $H(j\omega)$  represents the receptance (FRF) of the system. Using Eq. (3), the peak value from the FRF curve is obtained to find the resonant frequency of the system. By plotting the real vs. imaginary values of the system near the resonant zone, a circle can be fit, which is called as Nyquist plot [32]. This technique is used to calculate the damping ratio of the system. Eq. (4) is used to compute the damping ratio of the system by feeding the parameters obtained from the Nyquist plot. Here, the angles made by the arc subtended between points on either side of the natural frequency are represented by  $\theta_x$  and  $\theta_y$ .

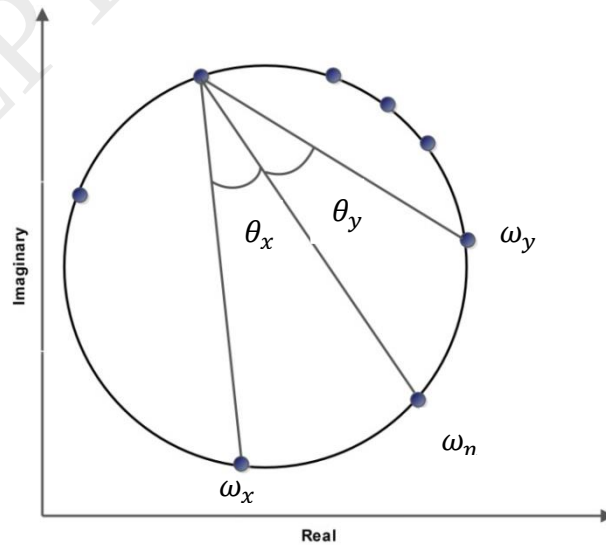


$\omega_x$  and  $\omega_y$  are the corresponding frequencies of the arc subtended, as shown in Fig.7.  $\omega_n$  represents the natural frequency for which the damping ratio is to be computed. Points on the circumference of the circle depict the coordinates of real and imaginary components. Damping of each specimen is computed using the parameters obtained from the Nyquist circle, which is shown in Fig.7.



**Fig. 6** GUI for conducting the modal analysis using the DEWE soft data acquisition system.

$$\zeta = \frac{\omega_x^2 - \omega_y^2}{\omega_n^2 \left( \tan\left(\frac{\theta_x}{2}\right) + \tan\left(\frac{\theta_y}{2}\right) \right)} \quad (4)$$



**Fig. 7** Nyquist circle plot used to compute damping ratio in this study.

## 4. Results and Discussion

In this section, the results obtained from FEM and experimental studies are presented and discussed.

### 4.1 FEM results

Sandwich beams were modeled using a CAD modeling software Autodesk Inventor with overall dimensions of  $160 \times 25 \times 6 \text{ mm}^3$ . They include three core shapes (triangle, square, and hexagon) oriented in three different directions ( $0^\circ$ ,  $45^\circ$ , and  $90^\circ$ ). The face sheets of the beams were maintained to be of 0.6 mm thick. A sectional view of the modeled triangular structure with  $0^\circ$ -orientated core is shown in Fig. 8(a). Created models were imported to ANSYS workbench simulation software wherein modal analysis was carried out by fixing one end and allowing the other end to vibrate freely. Table 5 depicts the material properties employed in the simulation study. A tetrahedral element is chosen to generate the overall stiffness matrix of the sandwich beam as this element has computation capabilities of handling plastic materials. The results obtained from the modal analysis are presented in Table 6. Note that the stiffness in the transverse direction of the beams was higher than that in bending because the moment of inertia was dominant in the transverse direction.

**Table 5** Material properties of ABS material

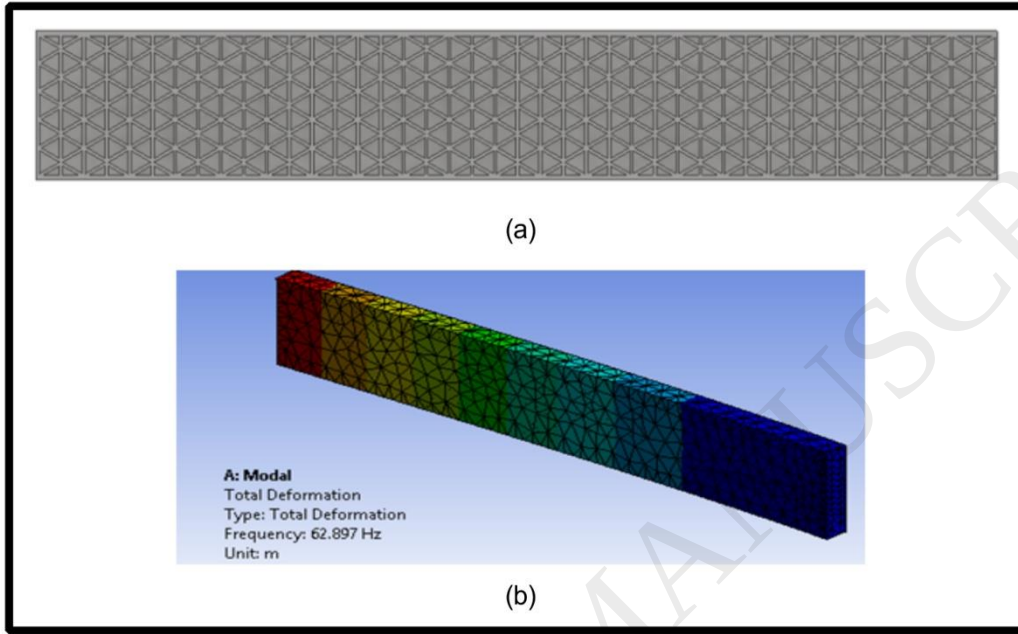
| Sl. No. | Material Properties | Values                 |
|---------|---------------------|------------------------|
| 1       | Density             | 1040 kg/m <sup>3</sup> |
| 2       | Poisson's Ratio     | 0.45                   |
| 3       | Young's Modulus     | 2.25 GPa               |

**Table 6** Bending resonant frequency obtained using a simulation study

| Shapes   | Orientation | Natural Freq (Hz) |
|----------|-------------|-------------------|
| Triangle | $0^\circ$   | 63.87             |
|          | $45^\circ$  | 69.80             |
|          | $90^\circ$  | 63.80             |
| Square   | $0^\circ$   | 72.7              |
|          | $45^\circ$  | 63.37             |
|          | $90^\circ$  | 72.7              |
| Hexagon  | $0^\circ$   | 72.00             |
|          | $45^\circ$  | 69.50             |
|          | $90^\circ$  | 60.85             |

Hence, the first natural frequency obtained by the simulation study is considered to be the bending resonant frequency only. From Table 6, it can be inferred that a sandwich structure with square core oriented at  $0^\circ$  and the  $90^\circ$  provided the highest natural frequency

among other specimens, whereas the hexagon core with  $90^\circ$  orientations provided the lowest value. Furthermore, it can be observed from Table 6 that the hexagon and square core beams orientated at  $0^\circ$  portrays nearly the same resonant frequency. In the above context, the natural frequency can be correlated with the stiffness of the structure, because the mass of the specimens is maintained the same for each sample. Explanation of the achieved variation in stiffness is revealed in the following section.



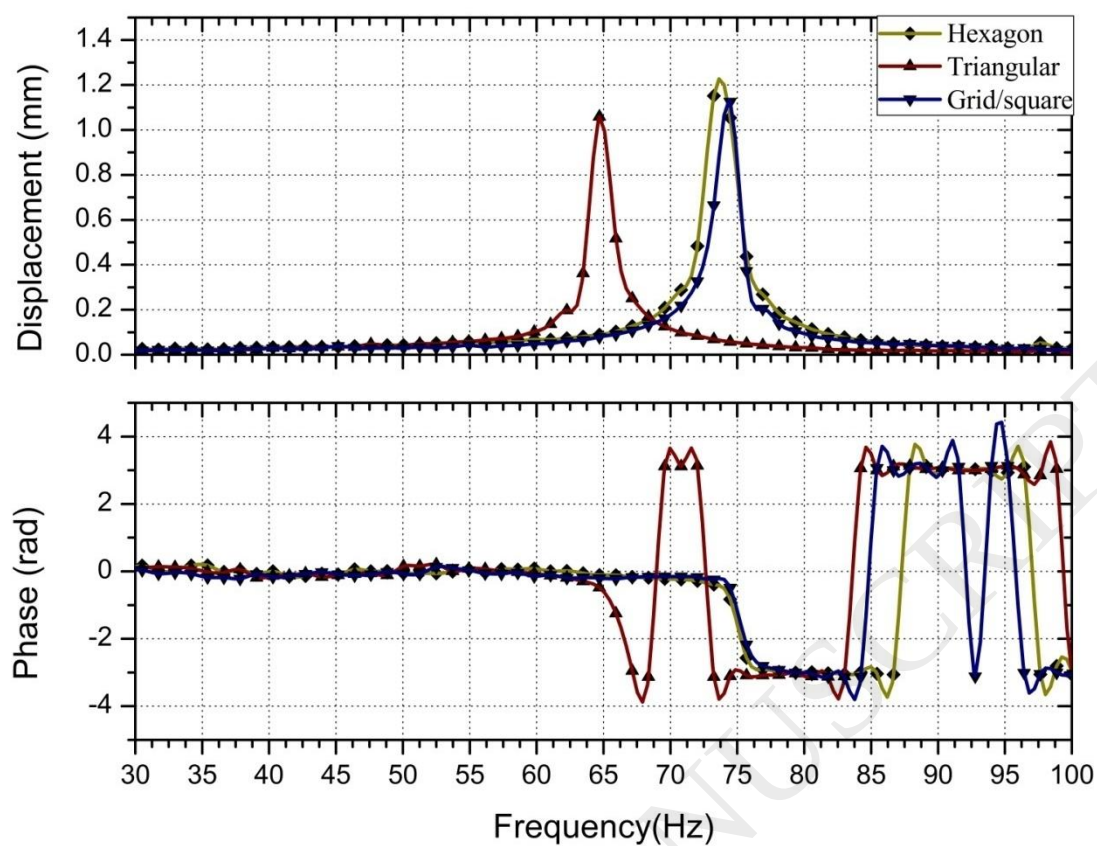
**Fig. 8** Modeled sandwich beam and typical FEM result of triangle core shape at  $0^\circ$  orientation (a) Sectional view of 3D model, (b) FEM results

## 4.2. Experimental Results

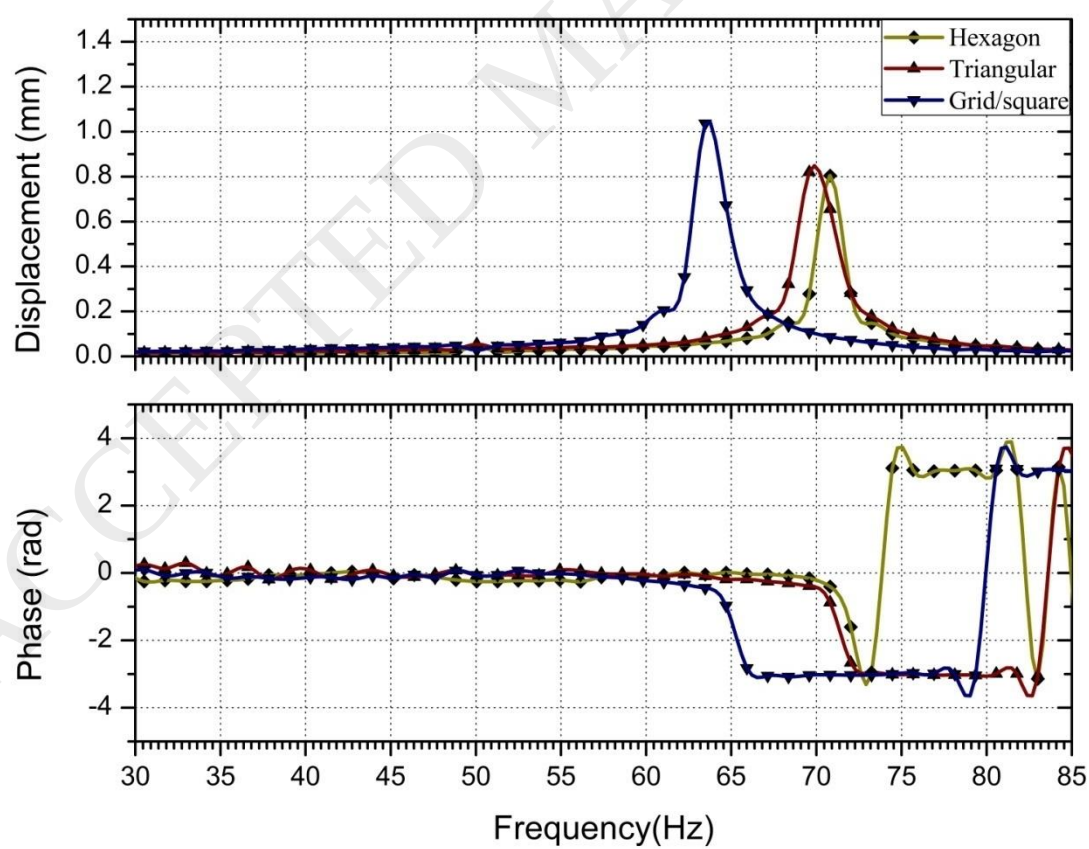
### 4.2.1. Comparison of stiffness:

To validate the results presented using simulation studies, the resonant frequency of the sandwich beams was computed experimentally. In the experimental setup, a similar boundary condition was attained as in the simulation studies. A rigid fixture was used to clamp one end of the beam, and roving hammer impact testing was conducted to obtain the FRF of the specimen. A soft tip was used for exciting the system to determine the lower natural frequencies accurately. The entire beam was apportioned into five equidistant node points on which the impulse input was imparted. The response was measured using a noncontact-type displacement measurement sensor (laser-pickup). A typical output obtained from a sandwich structure having square core shape is shown in Fig. 6. It depicts five curves acquired from five different hits; this indicates that the fundamental frequency obtained for all the runs was the same.

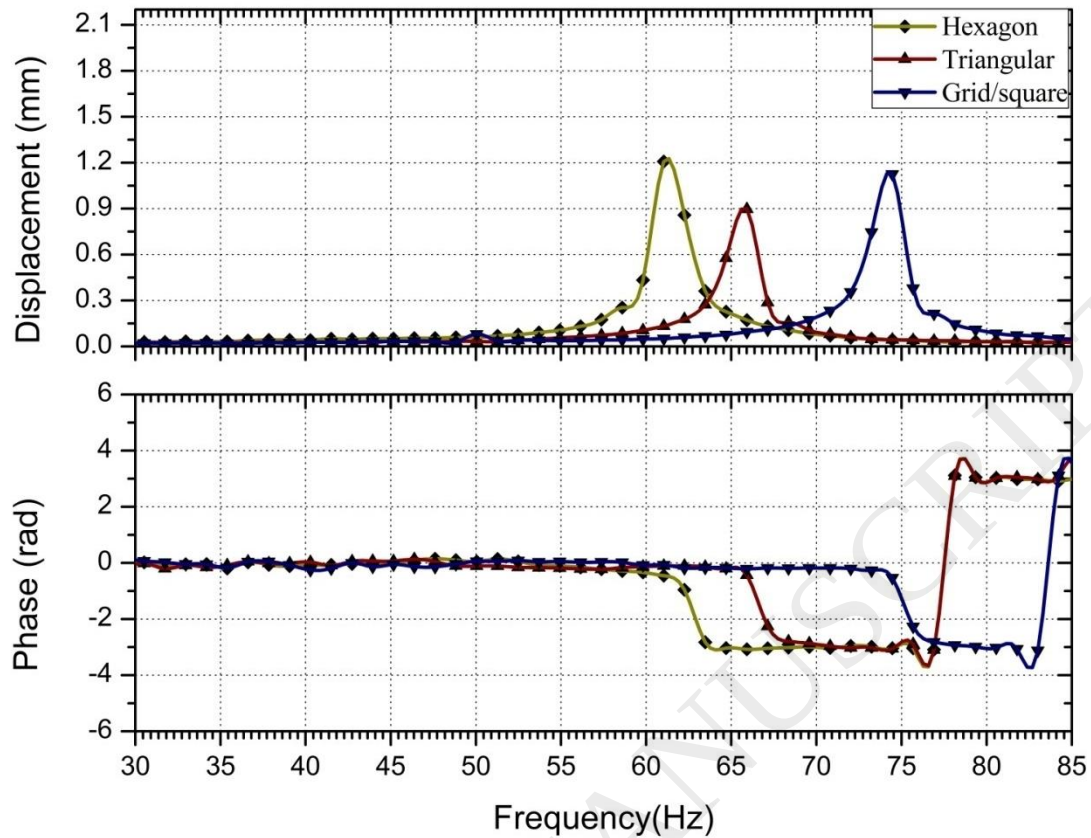




9(a)



9(b)



9(c)

**Fig. 9** Frequency Response Function of the beam having cores orientated in (a)  $0^\circ$ , (b)  $45^\circ$ , (c)  $90^\circ$

Comparison of the bending resonant frequency and phase values of different core shapes oriented at  $0^\circ$ ,  $45^\circ$ , and  $90^\circ$  directions are shown in Figs. 9(a-c), respectively. It can be observed from the figures that the phase values descend instantly after crossing a specific frequency; this confirms that the frequency at which the peak amplitude is attained from the FRF represents the natural frequency of the system.

Fig. 9(a) shows that hexagonal and square-shaped core structures exhibit the highest natural frequency, while the structure with triangular core provided the lowest natural frequency. From Fig. 9(b), it can be inspected that the hexagon core structure has the highest natural frequency, whereas the square core structure rendered the least. Similarly, from Fig. 9(c), it can be inferred that the square core structure confers the highest natural frequency, whereas the structure with hexagon core provided the least.

To explain the reason behind the variation in the stiffness with parameters such as the orientation of fiber, core shape, and the number of infill cores, the following procedure is followed for each specimen to understand the stiffness behavior precisely:

Step 1: Bending stiffness will be enhanced if the number of fibers oriented in the

longitudinal direction is more than that in the transverse direction. This behavior can be correlated to springs connected in parallel, where the overall stiffness of the system is the summation of the stiffness of individual springs.

Step 2: If the fibers are oriented in a particular angle, then the bending stiffness will be distributed between longitudinal and transverse directions, and their magnitudes can be determined by resolving them into sine and cosine components.

Step 3: Because the space occupied by different core shapes is distinct, also to maintain the same mass for all the specimens, the number of infills was altered. Therefore, it is postulated that the percentage of infill is an important parameter to be considered while computing the bending stiffness

FRF obtained using the modal analysis is displayed in Fig. 9(a-c). From the acquired results, the following conclusions can be derived:

1. Table 2 shows that the square core shape orientated at  $0^\circ$  and  $90^\circ$  has the most number of fibers orientated in the longitudinal direction, followed by the hexagon one, and the least was observed in the case of triangular core shape. It can also be discerned that none of the fibers are entirely oriented in the longitudinal direction for triangular core shape beam, which in turn decreased its bending stiffness compared to those of the other two core shape beams (Fig. 9(a)). This validates step 1 in the procedure proposed to distinguish the bending stiffness from the orientation of the fibers in a specific direction. From Table 2, it can be observed that the number of shapes for a square infill is more than that for the hexagonal infill. This is due to the less space consumed by the square infill compared to that by the hexagon infill. This validates step 3 in the procedure.
2. Similarly, in the case of sandwich beams orientated at  $90^\circ$ , the square infill shows superior bending stiffness followed by the triangle and hexagon infills (from Fig. 9 (c)). The triangular infill has a higher bending stiffness than the hexagon infill due to the presence of fibers in the longitudinal direction while it is absent for the hexagon infill. Further, it is worth noting that the number of infill cores found in the triangular pattern is more than hexagon pattern, which also contributed to increase the bending stiffness.
3. By comparing the bending natural frequency of the sandwich beam having infill orientated at  $45^\circ$  (Fig. 9 (b)), it can be observed that the infill with hexagonal shape provided higher stiffness than the other infills. However, the difference was

perceived to be less. The highest value of the hexagon infill may be due to the presence of more number of infills present in a fixed space. Further, it is worth observing that in all the cases, infills are not oriented in the longitudinal or transverse direction. Therefore, such structures can be implemented wherein the loads act in multiple regions, i.e., both in transverse and longitudinal.

To summarize, the highest natural frequencies for triangular, square, and hexagon core shapes were observed at  $45^\circ$ ,  $0^\circ/90^\circ$ , and  $0^\circ$  orientations, respectively. This observation emphasizes that different core shapes must be oriented accordingly to achieve a high bending stiffness. The lowest natural frequency was observed for hexagon infill oriented at  $90^\circ$ . Therefore, it can be concluded that the orientation of core is also critical in achieving a high bending stiffness. In the case of triangular and square core shapes, the number of cores obtained was more than that in the case of hexagon core shape. This is the reason that the lowest stiffness beam is found in hexagon pattern because the number of core shapes is least in the hexagon sandwich structure than in any other core shape. Therefore, in the case of hexagon cores, the orientation of shapes is crucial in achieving a high bending stiffness. Hence, from the above discussion, it can be concluded that the natural frequency of the sandwich structures depends equally on the infill (i) shape, (ii) number, and (iii) orientation.

**Table 7** Natural frequencies of different cores

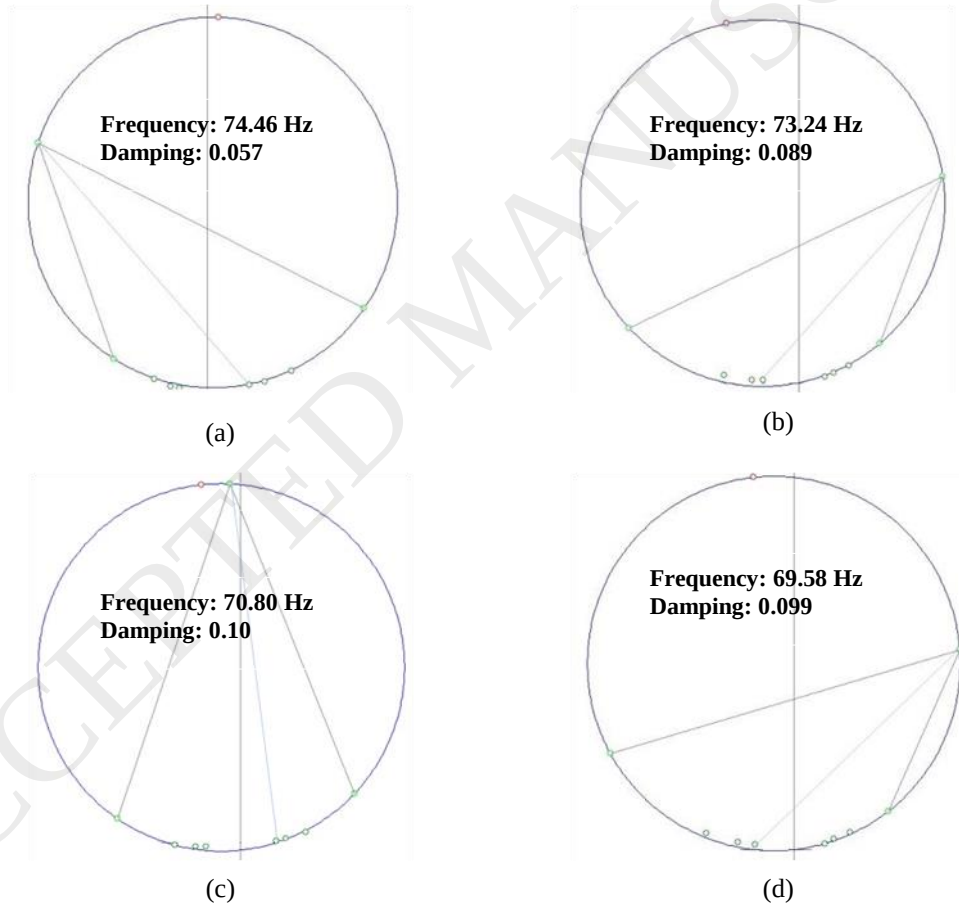
| Shape      | Orientation | Number of cores | Natural Freq (Hz) |       |           |
|------------|-------------|-----------------|-------------------|-------|-----------|
|            |             |                 | Expt.             | Theo. | Error (%) |
| Triangular | $0^\circ$   | 440             | 64.69             | 63.87 | 1.28      |
|            | $45^\circ$  | 484             | 69.58             | 69.80 | 0.31      |
|            | $90^\circ$  | 456             | 65.91             | 63.80 | 3.30      |
| Square     | $0^\circ$   | 720             | 74.46             | 72.7  | 2.42      |
|            | $45^\circ$  | 416             | 63.47             | 63.37 | 0.15      |
|            | $90^\circ$  | 720             | 74.46             | 72.7  | 2.42      |
| Hexagon    | $0^\circ$   | 333             | 73.24             | 72.00 | 1.72      |
|            | $45^\circ$  | 603             | 70.80             | 69.50 | 1.80      |
|            | $90^\circ$  | 384             | 61.03             | 60.85 | 0.29      |

#### 4.2.2. Comparison of damping ratio:

Damping is an important parameter to consider when designing the system because it helps in reducing the vibration amplitudes at resonant zones. In this study, the damping ratio

of top four specimens providing higher natural frequency is computed using the Nyquist circle plot, wherein real and imaginary components obtained from FRF are utilized to plot a circle.

The plots obtained from four sandwich structures having the highest stiffness, i.e., Square  $0^\circ$ , Hexagon  $0^\circ$ , Hexagon  $45^\circ$ , and Triangular  $45^\circ$ , are shown in Fig. 10 (a-d), respectively. Consolidated damping ratios, for all the structures, are tabulated in Table 8. From Table 8, it can be envisaged that the highest damping was observed for Triangular  $0^\circ$  followed by Hexagon  $45^\circ$ , Triangular  $45^\circ$ , and Square  $45^\circ$ . Damping ratio was found to be the minimum for Hexagon  $90^\circ$ . It can also be perceived from Table 8 that  $45^\circ$  orientation indicates the highest damping in each specimen. It can be further inferred from Tables 7 and 8 that the hexagonal shape oriented at  $45^\circ$  and  $0^\circ$  provides the highest stiffness and damping ratio.



**Fig. 10** Nyquist circle for (a) Square  $0^\circ$ , (b) Hexagon  $0^\circ$ , (c) Hexagon  $45^\circ$ , and (d) Triangular  $45^\circ$

**Table 8** Damping ratio obtained using the Nyquist plot

| Shape      | Orientation     | Number of cores | Damping Ratio |
|------------|-----------------|-----------------|---------------|
| Triangular | 0 <sup>0</sup>  | 440             | 0.102         |
|            | 45 <sup>0</sup> | 484             | 0.099         |
|            | 90 <sup>0</sup> | 456             | 0.032         |
| Square     | 0 <sup>0</sup>  | 720             | 0.057         |
|            | 45 <sup>0</sup> | 416             | 0.091         |
|            | 90 <sup>0</sup> | 720             | 0.057         |
| Hexagon    | 0 <sup>0</sup>  | 333             | 0.089         |
|            | 45 <sup>0</sup> | 603             | 0.10          |
|            | 90 <sup>0</sup> | 384             | 0.025         |

## 5. Conclusion:

The vibration response of sandwich structures with different in-fills patterned in the triangular, square, and hexagon shapes was studied in detail. Each infill pattern was analyzed at 0<sup>0</sup>, 45<sup>0</sup>, and 90<sup>0</sup> orientations. Nine specimens were modelled using FDM and modal analysis using FEM. Impact hammer tests experimentally validated the simulation test results. The natural frequency and damping ratio of the beam were experimentally estimated using the “DEWE soft data acquisition system”. This study primarily focused on the variation in stiffness and damping ratio caused by the changes in infill shapes even while retaining the mass, thickness of face sheet, and the outer dimension of the beam. From the simulation and experimental results, the following conclusions were derived:

1. Sandwich structure with hexagon core shape orientated at 0<sup>0</sup> and square core shape oriented at 0<sup>0</sup> and 90<sup>0</sup> provided higher bending stiffness as the natural frequencies were found to be the highest, while hexagon core with 90<sup>0</sup> orientation provided the lowest bending stiffness.
2. The stiffness of the sandwich beam structures depends equally on the infill (i) shapes, (ii) numbers, and (iii) orientations.
3. Irrespective of the shapes, the damping ratio was found to be the maximum at 45<sup>0</sup> orientation.
4. Damping ratio was found to be the maximum for triangular core shape oriented at 0<sup>0</sup> and the minimum for hexagon core shape at 90<sup>0</sup>. Of the two highest stiffness-providing shapes, i.e., hexagon and square patterns, the hexagon pattern showed a higher damping ratio ; hence, it can be utilized in different applications, to take the advantage of stiffness and damping.



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